

Ernest Sebastian - Section 2 Lift & Drag Kitten Assignment

28th July 2023

1 Given the following for a Plane travelling from LL to Kolkata

$$\text{Mass of Plane (M)} = 10,000 \text{ kg}$$

$$C_L = 1.1$$

$$\text{Surface Area} = 40 \text{ m}^2$$

$$\rho = 1.22 \text{ kg/m}^3$$

$$\text{Distance (D)} = 4325 \text{ km}$$

a) Calculate speed of the airplane

first we need to calculate the lift force then we can calculate the velocity of the plane.

$$L = \frac{(C_L \times \rho \times S \times V^2)}{2} = \sqrt{2(98100 \text{ N})}$$

$$1.1 \times 1.22 \text{ kg/m}^3 \times 40 \text{ m}^2$$

$$2L = C_L \times \rho \times S \times V^2$$

$$= \sqrt{196200 \text{ N} / 53.68 \text{ kg/m}^3}$$

$$V^2 = 2L / C_L \times \rho \times S$$

$$V = \sqrt{2L / (C_L \times \rho \times S)}$$

$$= \sqrt{3654.99 \text{ m/s}}$$

$$\text{Lift force} = m \times g$$

$$= 10000 \times 9.81 \text{ m/s}^2$$

$$= 98100 \text{ N}$$

$$= 60.5 \text{ m/s} \quad \text{ans.}$$

$$V = \sqrt{2L / (C_L \times \rho \times S)} \quad \nearrow$$

b) How long will it take to cover 4325 km @ 60.5 m/s

$$\text{Time} = \text{Distance} / \text{Speed}$$

$$T = D / S$$

$$T = \frac{4325000}{60.5 \text{ m/s}}$$

$$= \underline{\underline{20.02 \text{ Hrs}}} \text{ ans.}$$

c) Time it will take if the aircraft is being faced by 8 m/s headwind

$$\text{Speed} = 60.5 \text{ m/s} - 8 \text{ m/s} = 52.5 \text{ m/s}$$

$$\text{Distance} = 4325000 \text{ m}$$

$$\text{Time} = \frac{4325000 \text{ m}}{52.5 \text{ m/s}}$$

$$= 82381 \text{ s}$$

$$= \underline{\underline{22.9 \text{ Hrs}}} \text{ ans.}$$

2. After refuelling the plane lands to Sweden

$$\text{Mass} = 12,500 \text{ kg}$$

$$\text{Frontal Area} = 17.5 \text{ m}^2$$

$$\text{Surface Area} = 40 \text{ m}^2$$

$$\text{Density} = 1.22 \text{ kg/m}^3$$

$$\text{Distance} = 6175 \text{ km}$$

a) Calculate Speed

Lift formula

$$L = \frac{C_L \times \rho \times S \times U^2}{2}$$

$$U = \sqrt{2L / (C_L \times \rho \times S)}$$

$$L = m \times g$$

$$= 12500 \text{ kg} \times 9.81 \text{ m/s}^2$$

$$U = \sqrt{2L / (C_L \times \rho \times S)}$$

$$= \sqrt{2(122625 \text{ N}) / (1.1 \times 1.22 \text{ kg/m}^3 \times 40 \text{ m}^2)}$$

$$= \sqrt{4568.7 \text{ m/s}}$$

$$= 67.6 \text{ m/s} \text{ ans}$$

b) Find the thrust if $C_D = 0.15$

$$\text{Drag Force} = \frac{C_D \times \rho \times A \times V^2}{2}$$

Drag equals thrust when travelling linearly

$$\text{Thrust} = \frac{(0.15 \times 1.22 \text{ kg/m}^3 \times (67.6 \text{ m/s})^2)}{2}$$

$$= 60.08568 \text{ kg} \times 9.81 \text{ m/s}^2$$

$$= \underline{\underline{589.4 \text{ N}}} \text{ ans}$$

c) Calculate power used

$$\begin{aligned} \text{Power} &= \text{Thrust} \times \text{Speed} \\ &= 589.4 \text{ N} \times 67.6 \text{ m/s} \\ &= 39872.64 \text{ W} \\ &= 39.872 \text{ kW} \end{aligned}$$